

ALTOS® Lite Loose Tube, Gel-Free, Single-Jacket, Single-Armored Cable 12 F, 50 µm multimode (OM3)

CORNING

Part Number: 012TUC-T4180D20

Corning ALTOS® Lite gel-free, single-jacket, single-armored cables are designed for campus backbones in direct-buried installations. The loose tube design provides stable and highly reliable transmission parameters for a variety of voice, data, video and imaging applications. These cables also provide high-fiber density within a given cable diameter while allowing flexibility to suit many system configurations. The single armored construction provides additional crush and rodent protection with a high-strength ripcord under the armor for easy stripping. Gel-free means the cables are fully waterblocked using craft-friendly, water-swappable materials which make cable access simple and require no clean up. The flexible, craft-friendly buffer tubes are easy to route in closures, and the SZ-stranded, loose tube design isolates fibers from installation and environmental rigors while allowing easy mid-span access. These cables have a medium density polyethylene jacket that is rugged, durable and easy to strip.

Features and Benefits

Gel-free waterblocking technology

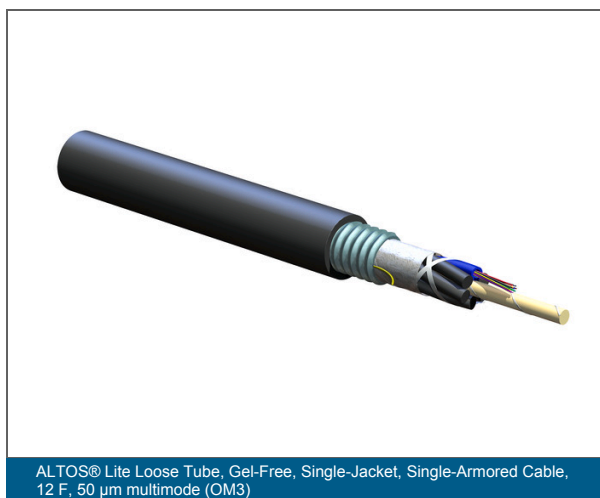
Craft-friendly cable preparation

Polyethylene jacket

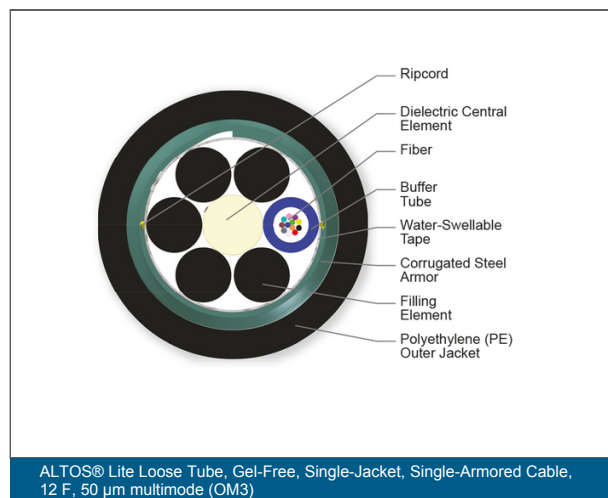
Rugged, durable and easy to strip (while providing superior protection against UV radiation, fungus, abrasion and other environmental factors)

Corrugated steel tape armor

Provides rodent resistance for direct-buried applications



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Specifications

General Specifications

Installation Methods	Aerial, Direct Buried, Duct
Cable Type	Loose Tube
Environment	Outdoor
Product Type	Steel Tape Armor
Fiber Category	50 µm MM (OM3)
Cable geometry	Round

Standards

RoHS	Free of hazardous substances according to RoHS 2011/65/EU
Common Installations	Outdoor lashed aerial, duct and direct-buried, indoor when installed according to National Electrical Code® (NEC®) Article 770
Design and Test Criteria	ANSI/ICEA S-87-640

Environmental Conditions

Notes	Corning recommends storing cable in a proper temperature environment prior to installation to allow the cable temperature to meet installation temperature range specifications for best installation results.
Temperature Range, Installation	-30 °C to 70 °C (-22 °F to 158 °F)
Temperature Range, Operation	-40 °C to 70 °C (-40 °F to 158 °F)
Temperature Range, Storage	-40 °C to 70 °C (-40 °F to 158 °F)

Cable Design

Central Element	Dielectric
Fiber Count	12
Number of Ripcords	2
Outer Jacket Color	Black

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Cable Design

Outer Jacket Material	Polyethylene (PE)
Tensile Strength Elements and/or Armoring - Layer 1	Corrugated steel tape armor
Buffer Tube Color	Blue
Buffer Tube Diameter	2.5 mm (0.1 in)
Number of Active Tubes	1
Number of Filling Elements	5
Number of Tube Positions	6
Tape	Water-swellaable
Fiber Coloring	Blue, Orange, Green, Brown, Slate, White, Red, Black, Yellow, Violet, Rose, Aqua
Fibers per Tube	12
Color Code Standards	Telcordia

Mechanical Specifications

Max. Tensile Strength, Long-Term	890 N (200.08 lbf)
Max. Tensile Strength, Short-Term	2700 N (606.98 lbf)
Nominal Outer Diameter	12.1 mm (0.48 in)
Min. Bend Diameter Installation	364 mm (14.33 in)
Min. Bend Diameter Operation	242 mm (9.53 in)

Optical Characteristics

Fiber Code	T
Performance Option Code	80
Fiber Category	OM3
Fiber Type	Multimode
Fiber Name	50 µm MM (OM3)
Maximum Attenuation	3.0 dB/km / 1.0 dB/km
Wavelengths	850 nm / 1300 nm
Fiber Compliance	IEC 60793-2-10

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Optical Characteristics

Fiber Core Diameter	50 µm
Serial 1 Gigabit Ethernet	1000 m / 600 m
Serial 10 Gigabit Ethernet	300 m / -
Minimum Effective Modal Bandwidth (EMB)	950 MHz*km / -
Min. Overfilled Launch (OFL) Bandwidth	1500 MHz*km / - 500 MHz*km

Dimensions

Cable Weight	129 kg/km (86.68 lb/1000 ft)
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